

KUSAL NIRMALA

Machine Learning Engineer — AI Undergraduate & Applied Researcher

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SUMMARY & EDUCATION

BSc (Hons) Artificial Intelligence undergraduate with hands-on experience spanning computer vision, embedded systems, and full-stack development. Proven leadership in hardware engineering and currently developing an original computer-vision research project applying pose estimation to martial arts technique analysis. Seeking Machine Learning Engineer roles with a long-term interest in research and academia.

BSc (Hons) Artificial Intelligence

University of Plymouth (UK) via NSBM

In Progress

Expected Graduation: 2029

BSc (Hons) Software Engineering (Top-Up)

Birmingham City University, UK

In Progress · Concurrent

Professional Higher Diploma in Software Engineering

SQA & UK Awards

In Progress

RESEARCH & LEADERSHIP

KaratePose_AI — Independent Research Project

In Progress

- Applying pose estimation (MediaPipe, YOLOv8-Pose) to karate techniques to benchmark accuracy degradation under high-velocity motion. Designing an event-triggered Extended Kalman Filter for joint tracking correction, structured as two formal manuscripts.

Team Lead — Team Hades

SLIOT Tournament, University of Moratuwa

- Led a student team to 2nd Runner-up at the University of Moratuwa SLIOT Tournament by designing and building a custom LoRa-based tracking system and PCB for fishery boats outside cellular coverage.

TECHNICAL PROJECTS

- **TaskMaster:** Developed a cross-platform mobile task management application utilizing a React Native frontend and a Java backend.
- **StaffMaster API:** Engineered a comprehensive REST API for staff management utilizing Java, Maven, and Jersey, complete with database integration.
- **Invoice Management System:** Built a scalable web-based invoice generator using Java, Servlets, and a REST API architecture.

SKILLS & AFFILIATIONS

Machine Learning & AI: Machine Learning, Deep Learning, Computer Vision, Pose Estimation

Software & Web: Python, Java, React Native, Node.js, PHP, Java Enterprise, SQL, Git

Hardware & IoT: PCB Design, LoRa, Embedded Systems

Affiliations & Extras: IEEE Member; Shotokan Karate Black Belt